



Analytical method for flow resistance of immiscible viscous fluid in heterogeneous porous media

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ABSTRACT

Fluid flow through porous media significantly impacts groundwater hydrology, carbon capture, utilization and storage (CCUS), as well as oilfield development. Due to limitations in analytical methods and an inadequate understanding of chemical flooding flow mechanisms, issues such as poor recovery efficiency and challenges in balanced reserve mobilization persist in heterogeneous reservoirs. Based on the disparities in water flooding stage in multi-layer reservoirs, this study integrates laboratory core flooding experiments, fractional flow theory, and equivalent flow resistance theory to establish a full-process flow resistance characterization model for multi-slug polymer flooding. This model accounts for factors such as polymer shear-thinning, adsorption and retention, and the coalescence, migration, and development of the oil bank. On this basis, this paper calculates the migration patterns and distribution profiles of oil, water, polymer, and the oil bank. The calculated results were compared and validated through core flooding experimental data, showing fitting accuracies of 98.02% for the fractional flow rate and 97.63% for the injection pressure. This model considers complex conditions and achieves high computational accuracy, which can contribute to the understanding of fluid flow mechanisms in chemical flooding for complex reservoirs and provide guidance for chemical flooding project design.

1. Introduction

The flow mechanisms of immiscible viscous fluids in heterogeneous porous media constitute a critical research focus across multiple disciplines, including hydrogeology (Schroeter et al., 2025; Cui et al., 2023), biomedicine (Rota et al., 2023), food industry (Tusek et al., 2021), nanoparticle synthesis (Sachdev et al., 2020) and petroleum engineering (Alfazazi et al., 2024; Chen et al., 2021). In petroleum engineering, the flow of oil and water is a typical example (Bazarin et al., 2025). Variations in pore geometry of porous media, properties of injected fluids, injection rates, and other factors significantly influence oil–water flow behavior and ultimate oil recovery efficiency (Cao et al., 2024; Cao et al., 2025). Therefore, understanding fluid flow mechanisms remains a critical focus throughout reservoir development. Quantifying flow resistance is fundamental to addressing the challenge of balanced

reserve mobilization across different layers during reservoir development. Current research focused on revealing interlayer flow differences and guiding development adjustments in multi-layer reservoirs by establishing calculation methods for flow resistance in individual layers has primarily concentrated on the water flooding (Bazarin et al., 2025). Based on water flooding theory, physical experiments, and numerical simulation, various dynamic flow resistance models have been developed to quantify interlayer interference, optimize water injection strategies, and allocate production rates (Li et al., 2025; Wang et al., 2020; Xue et al., 2025; Arab et al., 2020; Wang et al., 2022). Some methods also involve the numerical solution of two-phase flow resistance. However, water flooding in most Chinese oilfields has reached the high or even ultra-high water-cut stage, where dynamic reservoir heterogeneity is intensified (Zhang et al., 2024). Continuing with water flooding alone makes it difficult to achieve efficient reservoir development (Chen et al., 2024).

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Nomenclature			
Symbols		$f''(s_w)$	Second derivative of water cut with respect to water saturation, representing the curvature of the fractional flow curve.
a	Permeability Reduction Factor	R_p	Flow Resistance Under Multi-Slug Polymer
A	Core Cross-Sectional Area	RF	Resistance Factor
b_i	Minimum Permeability Reduction Factor Among the Transiting Slugs	RRF	Residual Resistance Factor
D_{rp}	Retardation Coefficient	μ_{eff}	Effective Polymer Viscosity
f_p	Water Cut Corresponding to Water Saturation in the Oil-Polymer Relative Permeability Curve	μ_{pm}	Viscosity at the Inflection Point on the Viscosity-Concentration Curve
$f'(s_w)$	Slope of the Water Cut Curve at a Water Saturation	μ_o	Oil Viscosity
K	Absolute Permeability	μ_w	Water Viscosity
K_{ro}	Oil Phase Relative Permeability	S_{wf}	Front Water Saturation
K_{rw}	Water Phase Relative Permeability	S_{wm}	Maximum Water Saturation
K_{rp}	Polymer Relative Permeability	S_{we}	Outlet End Water Saturation
L	Core Length	S_{wc}	Irreducible Water Saturation
ΔP_p	Polymer Injection Pressure	S_{w1}^*	Polymer Front Water Saturation
ΔP_{pw}	Subsequent Water Injection Pressure	S_{w2}^*	Oil Bank Front Water Saturation
ΔP_w	Water Injection Pressure	x_f	Water Front Position
$Q(p)$	Cumulative Polymer Injection Volume / Polymer Injection Rate	x_{f0}	Water Front Position at the End of Water Flooding
R	Flow Resistance	x_{fop}	Oil Bank Front Position
$R1$	Flow Resistance of the Polymer	x_{fp}	Polymer Flooding Front Position
$R2$	Flow Resistance of the Oil Bank	Δx_{pi}	Injection Distance of Each Slug
$R3$	Flow Resistance of the Oil-Water	ϕ	Effective Porosity
$R4$	Flow Resistance of the Oil	ϕ_e	Inaccessible Porosity

Consequently, most oilfields have transitioned to chemical flooding (Li et al., 2025; Li et al., 2025; Beteta et al., 2025; Lu et al., 2021). The commonly used methods for evaluating the performance of chemical flooding include physical simulation methods (Li et al., 2025; Chen et al., 2025; Liu et al., 2025), numerical simulation methods (Ahangar Darabi and Ahmadiouydarab, 2024; Dong et al., 2024), and analytical methods (Hwang et al., 2019; Mohammadnejad et al., 2021; Wang et al., 2025). Although the results from physical simulation methods are authentic and reliable, the scope of investigation is discontinuous and limited, unable to simulate all reservoir types (Chen et al., 2025; Song et al., 2024). Furthermore, the findings typically provide insights into reservoir behavior at specific instances and specific locations during displacement, thus lacking dynamic continuity (Jin et al., 2024; Lai et al., 2020). Sherif Fakher fitted mathematical models for polymer injectivity concerning polymer concentration, injection rate, and average reservoir pore throat diameter using the results of microtube experiment (Fakher, 2020). Xu Ke et al. demonstrated through microscopic experiments that viscoelastic oscillation is a key mechanism for enhanced oil recovery by viscoelastic fluids (Xie et al., 2022). Hou Jian et al. investigated the applicable permeability gradient of polymer and heterogeneous fluid agents using dual-sandpack experiments and validated the EOR effect of chemical flooding through large-scale 2D physical model experiments (Wu et al., 2022; Wu et al., 2020). Numerical simulation methods boast a wide range of application and high computational accuracy, but they rely heavily on high-precision history matching and parameter design (Zhong et al., 2025; David et al., 2025; Mansour et al., 2025). Ebuka David et al., considering the shear-thinning rheological properties of polymer, utilized pore network modeling and the Carreau rheological model to study the pore-scale EOR impacts of wettability, flow rate, shear-thinning, and agent concentration (David et al., 2025). Ahmed Mansour et al. employed numerical simulation to investigate the implementation feasibility of WAP (Water-Alternating-Polymer) injection (Mansour et al., 2025).

However, characterizing flow resistance in chemical flooding remains a challenging research area. Some scholars have employed analytical or semi-analytical methods to study injectivity during

polymer flooding (Hwang et al., 2019; Mohammadnejad et al., 2021; Wang et al., 2025). L. W. Lake defined injectivity of an injection well as the ratio of injection rate to flow pressure difference, treating polymer and water as a single aqueous phase. He derived a calculation formula for the pressure drop after polymer injection under conditions of planar radial steady-state flow in a circular formation with one injection well, subsequently obtaining an analytical solution for injectivity (W-LAKE, 1989). Sun Liang et al., assuming all reservoir layers were in a high water-cut stage and considering the fluid saturation distribution characteristics during non-piston-like polymer displacement, established an analytical injectivity model for polymer flooding in multi-layer reservoirs (Sun et al., 2019). Cheng Jiecheng et al. established a theoretical model that considers only the single-phase flow of chemicals in heterogeneous reservoirs. Using this simplified model, they analyzed the blocking effect of chemical agent viscosity on preferential flow channels (Cheng et al., 2020). Shi Leiting and Ye Zhongbin et al. developed an analytical model for the polymer flooding in multi-layer reservoirs, incorporating the shear-thinning and adsorption/retention properties of the polymer, and analyzed the effects of polymer effective viscosity and residual resistance factor on oil displacement efficiency (Shi et al., 2019; Cao et al., 2022).

The mathematical models established by previous researchers are primarily based on non-Newtonian fluid flow theory and applied mathematical methods. Compared to conventional methods like dynamic analysis, laboratory experiments, field trials, and numerical simulation, they reduce data processing workload and simulation computation time. Furthermore, analytical solution methods can effectively reveal the flow mechanisms during polymer flooding. However, these existing mathematical models possess certain limitations. For instance, they do not account for the differences in water flooding stage in multi-layer reservoirs under commingled injection and production. They also fail to consider the dynamic distribution of fluid saturation between injectors and producers during non-piston-like polymer displacement. Additionally, these models cannot resolve the variation characteristics of injectivity in individual layers under multi-slug combinations.

In view of this, based on laboratory experimental results, this paper utilizes the Buckley-Leverett equation to account for the water flooding stages of multi-layer reservoirs. By applying polymer flooding fractional flow theory and the equivalent flow resistance method, and considering the superposition calculation of four parts—polymer, oil bank, oil–water two-phase, and pure oil phase—a dynamic flow resistance calculation model for multi-layer, multi-slug combined chemical flooding was derived. And the model's accuracy was validated using the fractional flow and injection pressure data from laboratory experiments.

2. Mathematical model

In multi-layer reservoirs with strong heterogeneity, the water flooding stages vary significantly between layers. Therefore, accurately describing the water flooding stages and the water front position in each layer is a prerequisite for conducting dynamic flow resistance calculations for multi-slug chemical flooding. To address this, this section first potential scenarios during the chemical flooding, arising from the differences in water flooding stage between layers, are discussed. Secondly, characterizes and predicts the flow resistance throughout the entire water flooding based on the Buckley-Leverett equation, obtaining the water flooding stage of each layer. Subsequently, using the method proposed by Jiang for calculating the oil bank front saturation and migration velocity, a flow resistance calculation model considering the superposition of four parts (oil, water, polymer, and oil bank) is established. Finally, based on the water-electricity similarity principle and multi-slug flow experiments, a flow resistance calculation model for the multi-slug chemical flooding in multi-layer reservoirs is proposed (see Fig. 1).

2.1. Analysis of potential scenarios during the chemical flooding

When transitioning to polymer flooding under different water flooding stages, the distribution of oil, water, and polymer varies. Based on the water flooding stages of each layer, the following three scenarios are discussed: (1) The layer is in the high water-cut stage, with water already produced at the outlet, as indicated by permeability K_3 in Fig. 2. (2) The layer is in the middle stage of water flooding, with the water front located in the middle of the core, and no water production yet, as indicated by permeability K_2 in Fig. 2. (3) The layer is in the early stage of water flooding, largely unswept due to interlayer heterogeneity during water flooding, as indicated by permeability K_1 in Fig. 2. The production performance during the chemical flooding stage will subsequently be analyzed case by case, based on these three water flooding scenarios.

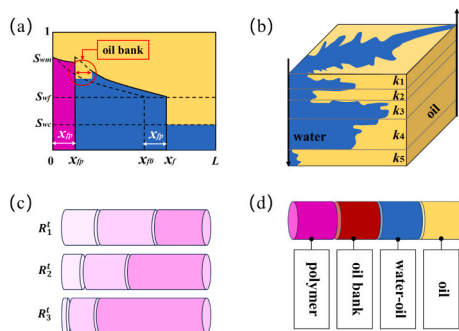


Fig. 1. Mind map of the analytical method (a) Potential scenarios during the chemical flooding. (b) Water flooding stages in heterogeneous reservoir. (c) Multi-slug polymer flow. (d) Superposition flow of four parts (oil, water, polymer, and oil bank).

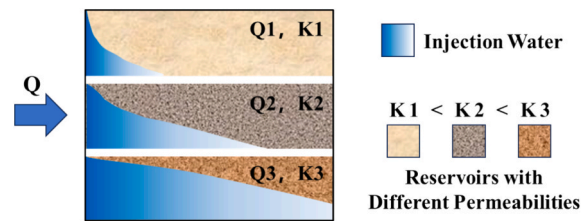


Fig. 2. Water flooding stages in a multi-layer strongly heterogeneous reservoir.

2.1.1. Scenario 1: Production dynamics during polymer flooding in the high water-cut stage

During commingled production in multi-layer reservoirs, the high-permeability layer has already entered a high water-cut stage. The oil–water distribution at the end of water flooding is shown in Fig. 3(a). Switching to chemical flooding under these conditions leads to the formation of an oil bank at the polymer front. As polymer injection proceeds, the following three distribution relationships of oil, water, polymer, and the oil bank are considered:

(1) Start of Polymer Injection: At the beginning of polymer injection, the oil bank gradually forms and expands. The flow resistance within the core comprises three parts: the polymer-oil two-phase region, the oil bank region, and the oil–water two-phase region, as shown in Fig. 3(b).

(2) Oil Bank Front Reaches Outlet: As displacement proceeds, the oil bank continues to grow, migrates, and is eventually displaced. When the oil bank front reaches the outlet, the flow resistance within the core consists of two parts: the polymer-oil two-phase region and the oil bank region, as shown in Fig. 3(c).

(3) Oil Bank is Fully Produced: With continued displacement, the oil bank is gradually produced. Once the oil bank is completely produced, the flow resistance within the core consists solely of the polymer-oil two-phase region, as shown in Fig. 3(d).

2.1.2. Scenario 2: Production dynamics during polymer flooding in the mid water flooding stage

For layers in the mid-stage of water flooding, where the water front has not reached the core outlet, the oil–water distribution at the end of water flooding is shown in Fig. 4(a). Switching to chemical flooding under these conditions leads to the following distribution scenarios for oil, water, polymer, and the oil bank:

(1) Start of Polymer Injection: As polymer injection begins, the oil bank gradually forms and expands, while the water front has not yet reached the core outlet. The flow resistance within the core comprises four parts: the polymer-oil two-phase region, the oil bank region, the oil–water two-phase region, and the oil phase region, as shown in Fig. 4(b).

(2) Water Front Reaches Outlet: As displacement proceeds, the water front reaches the core outlet. The flow resistance within the core now consists of three parts: the polymer-oil two-phase region, the oil bank region, and the oil–water two-phase region, as shown in Fig. 4(c). And then, the oil–water two-phase region is gradually produced. The flow resistance composition remains the same as in the previous stage but changes spatially: polymer-oil two-phase region, oil bank region, and oil–water two-phase region, as shown in Fig. 4(d).

(3) Oil Bank Front Reaches Outlet: With continued displacement, the oil bank is gradually produced, marking a period of high oil production rate during chemical flooding. The flow resistance within the core consists of two parts: the polymer-oil two-phase region and the oil bank region, as shown in Fig. 4(e).

(4) Oil Bank is Fully Produced: In the later stages, after the oil bank is completely produced, the flow resistance within the core consists solely of the polymer-oil two-phase region, as shown in Fig. 4(f).

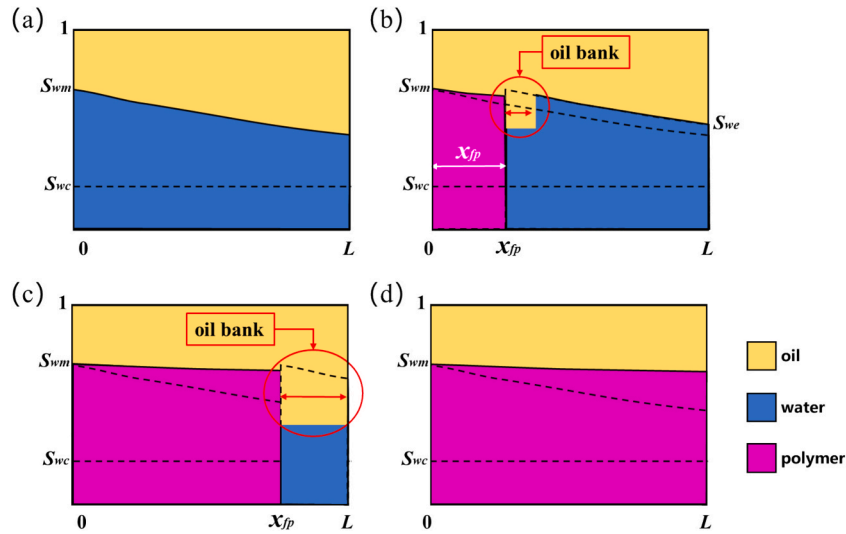


Fig. 3. Description of production dynamics during polymer flooding in the high water-cut stage (a) End of water flooding. (b) Start of polymer injection. (c) Oil bank front reaches outlet. (d) Polymer front reaches outlet.

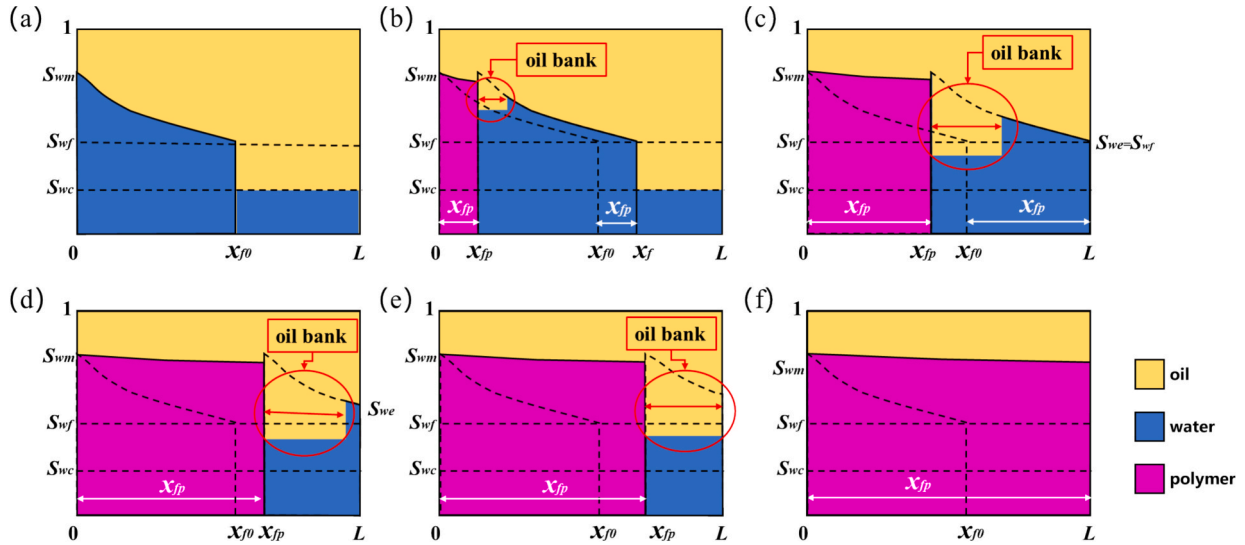


Fig. 4. Description of production dynamics during polymer flooding in the mid water flooding stage (a) End of water flooding. (b) Start of polymer injection. (c) Water front reaches outlet. (d) Water saturation at the outlet increases. (e) Oil bank front reaches outlet. (f) Polymer front reaches outlet.

2.1.3. Scenario 3: Production dynamics during polymer flooding in the early water flooding stage

During commingled production in multi-layer reservoirs, low-permeability layers, affected by permeability contrast, often exhibit poor water flooding and may be partially unswept, possibly only in the early stage of water flooding, as shown in Fig. 5(a). When switching to

chemical flooding under these conditions, the distribution relationships of oil, water, and polymer include two main scenarios:

(1) Start of Polymer Injection: At the beginning of polymer injection, the oil bank gradually forms and expands. The flow resistance within the core comprises four parts: the polymer-oil two-phase region, the oil bank region, the oil-water two-phase region, and the oil phase region, as

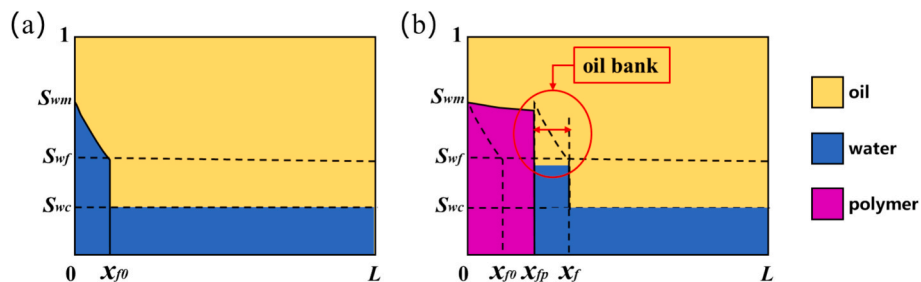


Fig. 5. Description of production dynamics during polymer flooding in the early water flooding stage. (a) End of water flooding. (b) Start of polymer injection.

analyzed for the mid-stage water flooding.

(2) Waterflood Front Submerged: Due to the limited extent of water flooding, the water front has not advanced deeply into the core. As polymer injection proceeds, the expanding oil bank's front might advance beyond the original water front position. This results in the flow resistance within the core consisting of only the polymer-oil two-phase region, the oil bank region, and the pure oil phase region, as shown in Fig. 5(b).

2.2. Flow resistance analytical model for the water flooding

The classical B-L equation is used to calculate the flow resistance during water flooding under different scenarios. Model assumptions: (1) Capillary and gravity forces are neglected. (2) Oil and water are incompressible. (3) Oil and water viscosities remain constant during displacement.

Before water breakthrough, where the water front has not reached the core outlet, the flow resistance calculation is divided into two parts: the resistance of the oil phase region and the oil–water two-phase region (Eq. (1) (Bazarin et al., 2025)).

$$R = \frac{u_o x_f}{KA f_{(S_{wf})}} \int_{S_{wm}}^{S_{wf}} \frac{f''(S_w)}{K_{rw} \mu_o / \mu_w + K_{ro}} dS_w + \frac{\mu_o (L - x_f)}{KA} \quad (1)$$

After water breakthrough at the core outlet, the water saturation at the core outlet changes continuously, and the resistance within the core consists only of the oil–water two-phase region (Eq. (2) (Bazarin et al., 2025)).

$$R = \frac{\mu_o L}{KA f_{(S_{we})}} \int_{S_{wm}}^{S_{we}} \frac{f''(S_w)}{K_{rw} \mu_o / \mu_w + K_{ro}} dS_w \quad (2)$$

2.3. Flow resistance analytical model for the polymer flooding

Given that the flow of different chemical agent in the core is influenced by multiple factors, certain simplifications are applied during model calculation. The following assumptions are made for this model:

(1) The viscosity of the polymer changes little during its migration after entering the core, and the effective viscosity μ_{eff} obtained from flow experiments is used in the model calculations.

(2) The effects of gravity and capillary force during displacement are neglected.

(3) Fluids are divided into water and oil phases; polymer is only soluble in water and insoluble in oil.

(4) Polymer only alters the relative permeability of the water phase and does not change the relative permeability of the oil phase (Zhong et al., 2020).

(5) Flow conforms to the generalized Darcy's law.

(6) Cross-flow between layers is not considered.

Specifically, the assumptions of “no crossflow” and “neglecting gravity and capillary forces” are interrelated. The former simplifies the 3D multi-layer system into multiple parallel 1D flow units, allowing the model to focus on dynamic flow resistance changes along the main flow direction. The latter further simplifies the driving force within each 1D unit to be dominated by viscous forces driven by the injection pressure gradient. This combination implies that the model does not currently account for fluid exchange between layers or complex saturation distributions caused by gravity segregation or capillary imbibition. This simplification clarifies the core mechanism of how polymer properties and slug sequences control interlayer fractional flow and is reasonable for the core-scale displacement and reservoir development conditions studied here, where the injection-production pressure difference far exceeds the contribution of gravity or capillary forces.

The method proposed by Jiang et al., which solves the continuity equations for water and polymer, is adopted here to determine the water saturation at the polymer solution front (Eq. (3)) and the water saturation

at the oil bank front (Eq. (4) (Sun et al., 2019)).

$$\left(\frac{df_p}{ds_w} \right)_{S_{w1}^*} = \left(\frac{f_p}{S_w + D_{rp} - \phi_e} \right)_{S_{w1}^*} \quad (3)$$

$$\left(\frac{f_p}{S_w + D_{rp} - \phi_e} \right)_{S_{w1}^*} = \frac{f_p(S_{w1}^*) - f_p(S_{w2}^*)}{S_{w1}^* - S_{w2}^*} \quad (4)$$

Using the method of characteristics and the graphical method, it is known that on the water cut versus water saturation curve, drawing a tangent from the point $(-D_{rp} + \phi_e, 0)$ to the polymer-oil fractional flow curve a tangent point. The water saturation corresponding to this tangent point is the water saturation at the polymer front, S_{w1}^* . The intersection point of this tangent line with the water–oil fractional flow curve corresponds to the water saturation at the oil bank front, S_{w2}^* , as shown in Fig. 6. Where D_{rp} is the dimensionless retardation coefficient and ϕ_e is the inaccessible pore volume, which are descriptive parameters for the adsorption retention and injectivity of polymers.

According to the polymer front advance equation, the migration velocity of the polymer front position x_{fp} is $f'(S_{w1}^*)$, and the migration velocity of the oil bank front position x_{fop} is $f'(S_{w2}^*)$.

By analyzing the distribution states of oil, water, and polymer during the chemical flooding under different water flooding stages, the flow resistance for the entire polymer flooding is solved segmentally, including four parts: polymer, oil bank, oil–water two-phase, and oil phase.

Polymer front position and corresponding flow resistance calculation shown in Eq. (5) ~ 6.

$$x_{fp} = \frac{Q(P)}{A\phi} f'(S_{w1}^*) \quad (5)$$

$$R1 = \frac{u_o x_{fp}}{KA f_{(S_{w1}^*)}} \int_{S_{wp}}^{S_{w1}^*} \frac{f''(S_w)}{K_{rp} \mu_o / \mu_{eff} + K_{ro}} dS_w \quad (6)$$

Oil bank front position and corresponding flow resistance calculation shown in Eq. (7) ~ 8.

$$x_{fop} = \frac{Q(P)}{A\phi} f'(S_{w2}^*) \quad (7)$$

$$R2 = \frac{u_o (x_{fop} - x_{fp})}{KA f_{(S_{w2}^*)}} \int_{S_{w1}^*}^{S_{w2}^*} \frac{f''(S_w)}{K_{rp} \mu_o / \mu_{pm} + K_{ro}} dS_w \quad (8)$$

Water flooding front position and corresponding flow resistance calculation shown in Eq. (9) ~ 10.

$$x_f = x_{fp} + x_{f1} \quad (9)$$

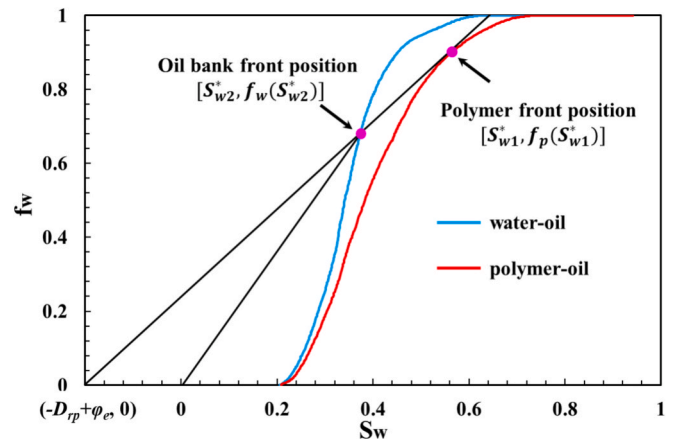


Fig. 6. Water-oil and polymer-oil fractional flow curves.

$$R3 = \frac{u_0(x_f - x_{fop})}{KA f'(s_{wf})} \int_{s_{w2}}^{s_{wf}} \frac{f''(s_w)}{K_{rw} \mu_o / \mu_w + K_{ro}} ds_w \quad (10)$$

Oil phase position and corresponding flow resistance calculation shown in Eq. (11) ~ 12.

$$x_o = L - x_f \quad (11)$$

$$R4 = \frac{\mu_o(L - x_f)}{KA} \quad (12)$$

2.4. Interactions during Multi-Slug polymer flow

The residual resistance factor is a coefficient reflecting reservoir damage considering the adsorption and retention of the polymer in the core. Based on its definition from the measurement method and Darcy's law:

$$RRF = \frac{\Delta P_{pw}}{\Delta P_w} = \frac{K}{aK} = \frac{1}{a} \quad (13)$$

The resistance factor is a comprehensive parameter considering the effects of shear thinning of the polymer in cores of different permeabilities and its adsorption/retention on the rock surface. Based on its test method and definition:

$$RF = \frac{\Delta P_p}{\Delta P_w} = \frac{K}{aK} \times \frac{\mu_{eff}}{\mu_w} = \frac{\mu_{eff}}{a} \quad (14)$$

The permeability reduction factor of the core and the effective viscosity of the polymer are obtained through resistance factor and residual resistance factor tests:

$$a = \frac{1}{RRF} \quad (15)$$

$$\mu_{eff} = \frac{RF}{RRF} \quad (16)$$

The change in flow resistance in the core during multi-slug polymer injection is considered for single-phase polymer flow. Model assumptions:

- (1) Before core displacement, the core is saturated with water with viscosity μ_w .
- (2) The viscosity of the polymer changes little during its migration in the core, maintained at μ_{eff} .
- (3) The effects of gravity and capillary force during displacement are neglected.
- (4) The effects of diffusion and dilution between multiple polymer slugs during flow are ignored.
- (5) Flow conforms to the generalized Darcy's law.

Considering the permeability damage caused by the adsorption and retention of multiple polymer slugs flowing through the core, the permeability reduction factor at each location in the core is selected as

the minimum permeability reduction factor b_i among the slugs that have passed through that location (Eq. (17)).

$$b_i = \min([a_j]_{j=1}^i) \quad (17)$$

Combined with the water-electricity similarity principle, the flow resistance of multiple slugs in the core is calculated by superposition (Eq. (18)), as shown in Fig. 7.

$$R_p = \frac{\sum \left(\left(\mu_{eff} \times \Delta x_{pi} \right) / (b_i) \right)}{K \times A} \left(x_p \geq L, \sum \Delta x_{pi} = L \right) \quad (18)$$

3. Model solution

3.1. Dynamic flow resistance calculation for the water flooding

- (1) Input the basic parameters of the core and fluids.
- (2) Calculate the initial flow resistance under oil-saturated conditions.
- (3) At each time step, allocate the flow rate based on the flow resistance.
- (4) Determine if the water front position in each layer has broken through the core outlet (water breakthrough).
- (5) If water breakthrough has occurred, the core contains only the oil-water two-phase region, and the flow resistance is calculated using Eq. (2). Conversely, the flow resistance consists of two parts: the oil-water two-phase region and the oil phase region, calculated using Eq. (1).
- (6) Determine if water flooding has ended. If water flooding has ended, output the results for the entire water flooding. If not, allocate the flow rate for the next time step based on the calculated flow resistance, and repeat steps (3) to (6), as shown in Fig. 8.

3.2. Dynamic flow resistance calculation for the Multi-Slug polymer flooding

Based on the differences in water flooding stage the layers, the displacement process of the oil bank is discussed case by case. This involves characterizing the distribution states of oil, water, and polymer during chemical flooding segmentally. Using the dynamic flow resistance characterization method for the entire chemical flooding, combined with the water-electricity similarity theory, the flow rate is allocated at each time step according to the flow resistance of each layer until the slug displacement concludes. The calculation flowchart is shown in Fig. 9.

- (1) Link the flow resistance of each core layer at the end of water flooding, and allocate the flow rate for each layer accordingly.
- (2) Determine if polymer is produced at the core outlet (polymer breakthrough).
- (3) If polymer breakthrough occurs, the flow resistance in the core is calculated considering only the polymer-oil two-phase region. If not,

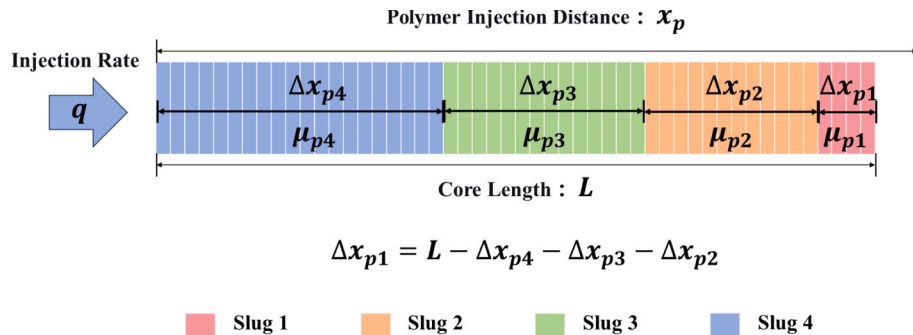


Fig. 7. Schematic diagram of flow resistance for four-slug polymer in the core.

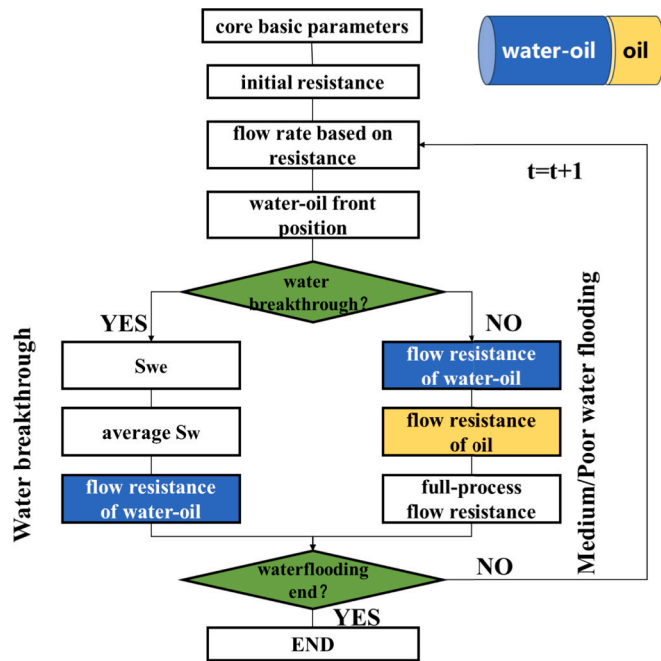


Fig. 8. Flowchart for calculating water flooding flow resistance in multi-layer reservoirs.

determine if the oil bank is produced at the core outlet.

(4) If the oil bank is produced at the core outlet, calculate the flow resistance by segmenting it into two parts: the polymer-oil two-phase region and the oil bank region. If not, determine if water is produced at the core outlet (water breakthrough).

(5) If water breakthrough has occurred at the core outlet, calculate

the flow resistance by segmenting it into three parts: the polymer-oil two-phase region, the oil bank region, and the oil–water two-phase region. If not, determine if the oil bank front has surpassed the water front position.

(6) If the oil bank front has surpassed the water front position, calculate the flow resistance by segmenting it into three parts: the polymer-oil two-phase region, the oil bank region, and the oil region.

(7) If the oil bank front has not surpassed the oil–water front position, calculate the flow resistance by segmenting it into four parts: the polymer-oil two-phase region, the oil bank region, the oil–water two-phase region, and the oil region.

(8) Based on the calculated flow resistance for each layer at this time step, allocate the flow rate. Repeat steps (2) to (7) until the cycle concludes.

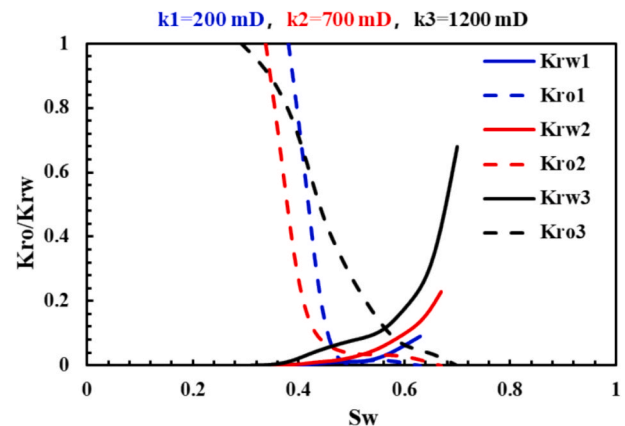


Fig. 10. Relative permeability curve.

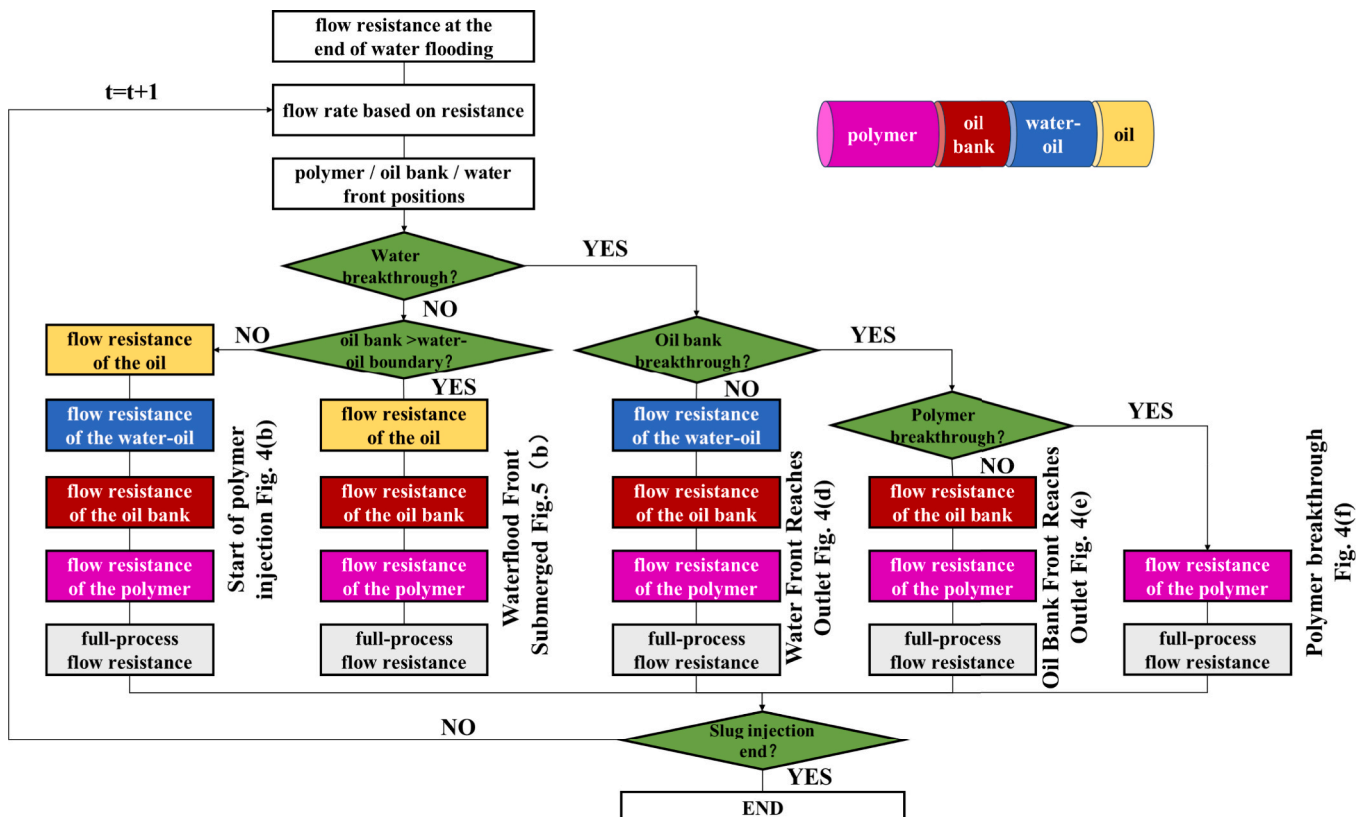


Fig. 9. Flowchart for calculating chemical flooding flow resistance in multi-layer reservoirs.

4. Experiment and model validation

The improved mathematical model was solved using MATLAB programming, and the results were compared with laboratory experimental data to validate the rationality of the model's solutions. First, multi-slug polymer flow experiments using three parallel cores were conducted to validate the flow resistance calculation model for multi-slug polymer flow. Second, the water flooding stage of each layer was analyzed using oil displacement experiments with three parallel cores combined with the B-L equation. Finally, the polymer flooding stage of the three-parallel-core oil displacement experiment was used to validate the rationality of the dynamic flow resistance calculation method for the polymer flooding in multi-layer reservoirs, and the production dynamics of each layer were analyzed.

Experimental conditions and basic model parameters are shown in Table 1: The cores were square long cores with dimensions of $4.5\text{ cm} \times 4.5\text{ cm} \times 30\text{ cm}$, characterized by high porosity and high permeability. The permeability combination was $1200\text{ mD} / 700\text{ mD} / 200\text{ mD}$, and the core porosity was approximately 0.30. The oil used was from Block QK with a viscosity of $12\text{ mPa}\cdot\text{s}$. The experimental water was injection water from the same block, with a salinity of 3213.04 mg/L . The polymer used was P01, with concentration of 3000, 1200, 800 mg/L. The resistance factor (RF) and residual resistance factor (RRF) for this polymer in cores of different permeabilities were measured, and the corresponding permeability reduction factor and effective viscosity results are shown in Table 2 below. Polymer solutions were prepared with injection water, aged for 24 h at $50\text{ }^{\circ}\text{C}$ to ensure full hydration and stable viscosity, and then filtered through a G3 sintered glass funnel prior to displacement. The experimental temperature was the reservoir temperature of the block, $50\text{ }^{\circ}\text{C}$. The displacement velocity was constant at $1.0\text{ cm}^3/\text{min}$.

4.1. Flow experiment and model validation

Flow experiments using three parallel cores were conducted with the three concentration mentioned above to validate the correctness of the flow resistance calculation model for multi-slug polymer flow. The porosities of the high, medium, and low permeability cores were 0.32, 0.29, and 0.28, respectively. The displacement scheme was: 0.7 PV of 1200 mg/L polymer + 0.45 PV of 3000 mg/L polymer + 0.45 PV of 800 mg/L polymer. The fractional flow and injection pressure during the experiment were recorded and compared with the model calculation results, as shown in the Fig. 11.

From Fig. 11 (a) and (b), it can be seen that the calculation results show a high degree of overlap with the experimental results. The model achieved R^2 values of 96.62% for the fractional flow rate and 96.98% for the injection pressure, calculated using all experimental data points. Furthermore, this algorithm can determine the fluid distribution within the core at different injection times, as shown in Fig. 11 (c).

After injecting the first slug (1200 mg/L), the fractional flow stabilized quickly. The injection pressure increased linearly at 0.305 MPa/PV , because the polymer rapidly compensated for permeability differences between layers. Since polymer resistance is much higher than water, the pressure rose linearly. When the injection volume reached 0.7 PV , the high-permeability core was filled with this slug, and polymer was just observed producing at the core outlet simultaneously. At this

Table 1
Summary of Model Input Parameters.

Parameter	Value	Parameter	Value
Layers	3	Water Viscosity	$1.0\text{ mPa}\cdot\text{s}$
Core Size	$4.5\text{ cm} \times 4.5\text{ cm} \times 30\text{ cm}$	Oil Viscosity	$12.0\text{ mPa}\cdot\text{s}$
Core Permeability	1200, 700, 200 mD	Displacement Velocity	$1.0\text{ cm}^3/\text{min}$
Core Porosity	0.31, 0.30, 0.287	Relative Permeability Information	See the Fig. 10

point, the injection distance ratio of the polymer in the high, medium, and low permeability cores was 30: 22: 11.5.

After the injection of the second slug (3000 mg/L), the core fractional flow remained relatively stable. This is because when this polymer entered the core, the flow resistance within the cores was in a relatively stable state, so the flow rate was allocated according to the needs of each layer. Moreover, the flow resistance of the second slug was much greater than that of the first slug, meaning the impact of the injected fluid on the flow resistance was far greater than that of the produced fluid, resulting in little change in the core fractional flow. For the same reason, the injection pressure increased linearly with the injection volume of the second slug at a rate of 1245 kPa/PV . When the total polymer injection volume reached 1.15 PV , at the end of the second slug injection, the injection distance ratio of the second slug in the high, medium, and low permeability cores was 18.9: 13.7: 7.3, and the remaining distance ratio of the first slug in the high, medium, and low permeability cores was 11.1: 16.3: 11.5. From this, it can be concluded that polymer was being produced from the medium and high permeability layers, while no polymer was observed yet from the low permeability layer.

After the injection of the third slug (800 mg/L), the fractional flow of the low-permeability core decreased slightly, while that of the medium and high-permeability cores increased slightly. The injection pressure decreased linearly at a rate of -167 kPa/PV as the third slug volume increased. This is analyzed to be primarily due to the injection of the third slug and the displacement of the first slug, but the difference in flow resistance between these two slugs within the cores was small, resulting in minimal change in fractional flow. When the cumulative polymer injection volume reached 1.4 PV , the second slug just reached the outlet of the high-permeability core. As it was gradually produced, the rate of injection pressure decrease changed to -1304 kPa/PV , and the fractional flow changes among the high, medium, and low permeability layers became significant. For such scenarios, the interlayer fractional flow can be predicted using this flow resistance calculation method. At this point, switching to an appropriate chemical slug—specifically, a slug with equivalent flow resistance to the produced fluid—can mitigate interlayer production disparities and achieve effective mobilization of the low-permeability layers.

This flow experiment demonstrates the accuracy of the resistance calculation method during multi-slug injection and shows that different slugs can adjust the fractional flow in heterogeneous cores during the injection process. It also indicates that this method can be used to adjust the injection system in real-time/timely based on reservoir resistance distribution. Oil displacement experiments and model validation will be conducted subsequently.

4.2. Oil displacement experiment and model validation

An oil displacement experiment using three parallel cores was conducted. After water flooding to a comprehensive water cut of 90%, 0.7 PV of polymer was injected, followed by water flooding until the comprehensive water cut reached 98%, at which point the experiment was stopped. Data on oil production, water production, and injection pressure were recorded during the experiment, and the results were compared with the model calculations. The polymer flooding scheme was: first inject 0.05 PV of 3000 mg/L polymer slug to utilize the water flooding channels and increase the flow resistance of the high-permeability layer, followed by injection of 1200 mg/L polymer slug. When the fractional flow of the high-permeability layer increased, a 0.05 PV high-viscosity polymer slug was injected. The final injected slug sequence at the end of the experiment was: $0.05\text{PV} + 0.35\text{PV} + 0.05\text{PV} + 0.15\text{PV} + 0.05\text{PV} + 0.05\text{PV}$.

From Fig. 12 (a) and (c), it can be seen that the calculation results show a high degree of consistency with the experimental results. The model achieved R^2 values of 98.02% for the fractional flow rate and 97.63% for the injection pressure, calculated using all experimental data points. Fig. 12 (b) shows the water cut changes at the outlet of each layer

Table 2
Polymer Physical Parameters.

Concentration / mg/L	Permeability / mD	Resistance Factor/ Dimensionless	Residual Resistance Factor/ Dimensionless	Permeability Reduction Factor/ Dimensionless	Effective Viscosity / mPa·s
3000	1200	168.3	2.4	0.42	70
	700	186.4	3	0.34	63
	200	195	3.5	0.28	55
1200	1200	36.5	1.5	0.66	24.1
	700	40.4	1.9	0.53	21.5
	200	42.3	2.2	0.46	19.5
800	1200	12.3	1	0.96	11.9
	700	13.7	1.2	0.8	11
	200	14.3	1.6	0.62	8.8

Note: Permeability Reduction Factor defines the extent to which absolute rock permeability is reduced following polymer adsorption retention. Effective Viscosity represents the equivalent Newtonian viscosity exhibited by the polymer solution flowing through porous media, comprehensively accounting for shear thinning effects, polymer molecular conformational changes, and pore geometry constraints (derived from Eqs. (15)–(16)).

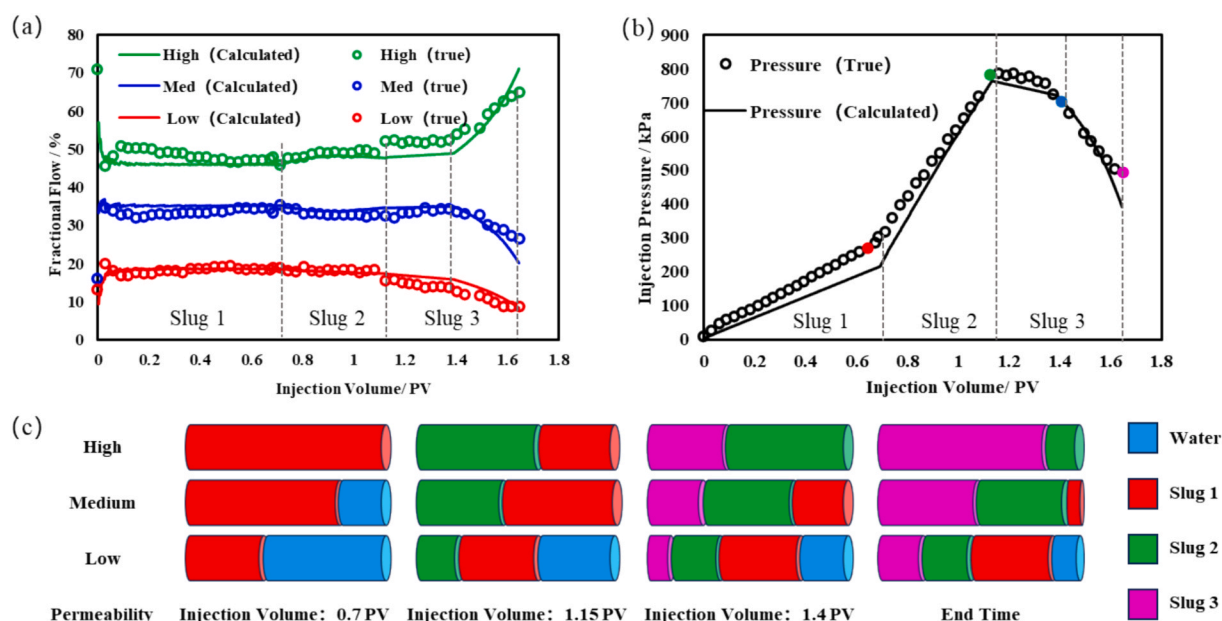


Fig. 11. Flow experiment model validation results (a) Fractional flow curve. (b) Injection pressure curve. (c) Fluid distribution in the cores ($K = 1200, 700, 200$ mD).

during the oil displacement process obtained from the experiment. Furthermore, this algorithm can determine the front positions and distribution states of various fluids in the cores at different injection times, as shown in Fig. 12 (d).

From Fig. 12 (b) and (d), it can be seen that during the water flooding stage, both the high and medium permeability layers experienced water breakthrough, and the water front positions matched the water breakthrough times. Water flooding ended at 0.58 PV injected; at this time, the water front in the low-permeability layer had reached 29.3 cm, and its water breakthrough time was just after switching to polymer injection, at a cumulative fluid injection of 0.62 PV. The consistency between these confirms the accuracy of the water flooding stage description. During the polymer injection stage, the oil bank formed rapidly, and the migration velocity of its front or the growth rate of the oil bank was very rapid, with negligible influence from fractional flow. The times when the oil bank front reached the core outlet in the high and medium permeability layers closely corresponded to the times of water cut decrease, further verifying the accuracy of the model calculation. In the low-permeability layer, influenced by the fractional flow, the polymer front never reached the core outlet.

From Fig. 12 (c), it can be seen that during the injection of the high-viscosity slug (3000 mg/L), the pressure increased rapidly and linearly at a rate of 8680 kPa/PV. This is because the resistance generated by this slug in the core was far greater than that of other fluids. This rate of

increase was higher than in the flow experiment, which is analyzed to be due to the additional resistance effect of the oil bank in this experiment.

During the injection of the medium-viscosity slug (1200 mg/L), influenced by the preceding slug and the oil bank, the pressure no longer showed a linear trend. Analyzing Fig. 12 (c) and (d) together, from 0.6 to 1.0 PV, the rate of injection pressure increase gradually slowed, with an average rate of 252 kPa/PV. The reason is believed to be that in the early injection period, both the injected fluid and the developing oil bank contributed to the increase in core resistance, causing a pressure increase rate higher than in the flow experiment. In the subsequent stage, the gradual production of the oil bank weakened the pressure increase rate, making it lower than the pressure increase rate in the flow experiment.

At 1.01 PV, the injection pressure dropped abruptly. Analysis combined with Fig. 12 (d) suggests that the initially injected high-viscosity slug (3000 mg/L) broke through to the core outlet. The injected fluid could not quickly compensate for this “gap,” leading to a rapid pressure drop. Fig. 12 (a) shows a rapid increase in the fractional flow of the high-permeability layer. An attempt was made to inject high-viscosity fluid at this point to improve the fractional flow of the medium and low permeability layers. However, lacking the dual regulatory effect of the oil bank, injecting the same 0.05 PV of high-viscosity fluid had a far less effect compared to the initial injection. After 1.2 PV, analysis combined with Fig. 12 (d) indicates that due to the decrease in injected fluid

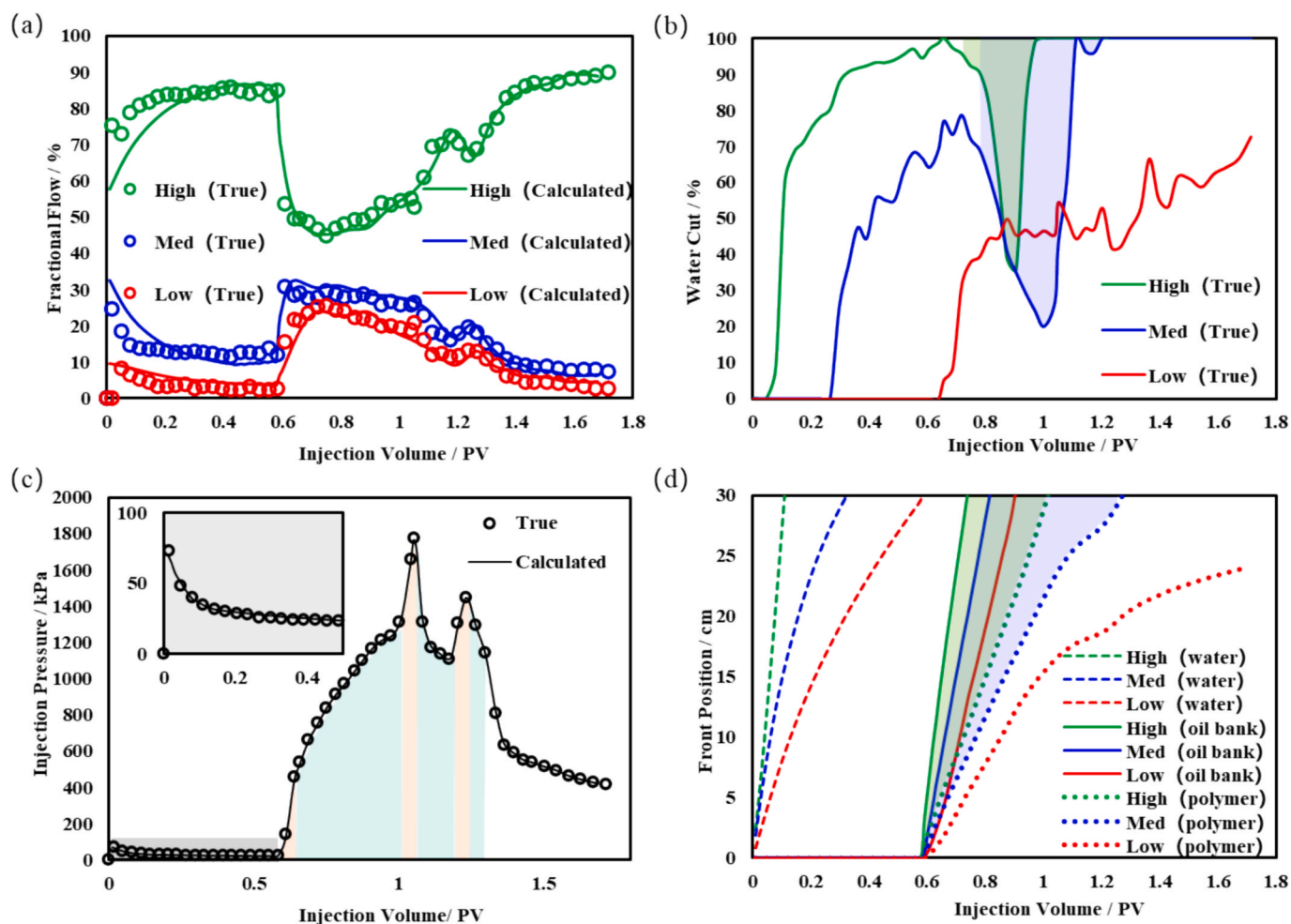


Fig. 12. Oil displacement experiment model validation results (a) Fractional flow curve. (b) Water cut curve. (c) Injection pressure curve. (d) Fluid front positions in the cores.

viscosity and the production of the oil bank, the injection pressure decreased rapidly under this “dual” effect.

Based on the above experimental results, control can be applied based on the resistance of the high-permeability layer: it is necessary to “replenish” corresponding resistance in the high-permeability layer timely or in a timely manner according to fluid production. Meanwhile, the oil bank plays a role in resistance regulation during its migration. The properties of subsequently injected fluids need to consider the changes in layer resistance caused by the development, growth, migration, and production of the oil bank. Minor discrepancies between model and experiments may arise from simplifications such as neglecting polymer degradation over time, slight variations in core heterogeneity not captured in the model, or measurement uncertainties in pressure sensors.

This model enables the calculation of water saturation at different locations along the core. The calculation results are shown in the Fig. 13. The blue curve represents the water flooding stage, the green curve represents the stage involving the oil bank, and the red curve represents the period after the polymer front reaches the core outlet. These results provide a clear visualization of the migration patterns of the four parts—oil, oil-water, oil bank, and polymer—during the displacement process.

Observing the blue curve, it is evident that the water front in the high-permeability layer reached the core outlet at 0.11 PV, which aligns closely with its water breakthrough time.

At this point, the water front positions in the medium and low-permeability layers were at 15.8 cm and 8.7 cm, respectively. When

the cumulative injection reached 0.32 PV, the water front in the medium-permeability layer arrived at the core outlet, also consistent with its water breakthrough time. Meanwhile, the water front in the low-permeability layer reached at 20.1 cm, near the core's mid-section. By the end of water flooding, the water front in the low-permeability layer was only 0.7 cm from the outlet, again matching its water breakthrough time well. Additionally, as water flooding and the medium and high-permeability layers experienced water breakthrough, the water cut at the core outlet continued to rise.

The green curve illustrates the development and migration of the oil bank within the core. Due to the higher fractional flow in the high-permeability layer and the lower remaining oil saturation after water flooding, the oil bank in this layer developed rapidly. By a cumulative injection of 0.7 PV, the oil bank in the high-permeability layer had reached a length of 16.4 cm, migrating to the mid-point of the core (half its total length). At this stage, the oil bank lengths in the medium and low-permeability layers were 9.4 cm and 5.7 cm, respectively. When the cumulative injection reached 0.74 PV, the polymer migration distances in the high, medium, and low-permeability layers were 11.7 cm, 9.1 cm, and 5.9 cm, respectively. The oil bank in the high-permeability layer achieved its maximum length of 20.5 cm, with its front reaching the core outlet, while the oil banks in the medium and low-permeability layers continued to grow, reaching lengths of 12.5 cm and 8.3 cm, respectively.

As displacement continued, at a cumulative injection of 0.81 PV, the oil bank in the medium-permeability layer reached its maximum length of 18 cm, with its front arriving at the core outlet. Combined with the production of the oil bank from the high-permeability layer, this period

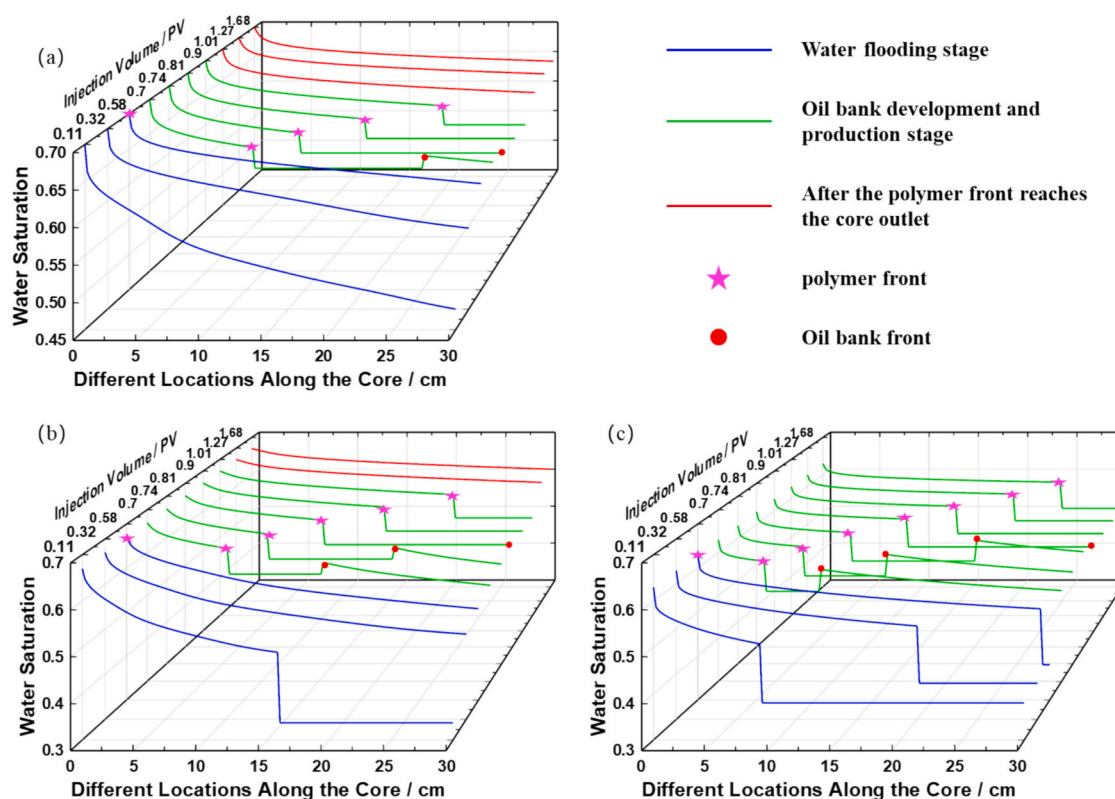


Fig. 13. Water saturation distribution at different locations along the core (a) $K = 1200$ mD. (b) $K = 700$ mD. (c) $K = 200$ mD.

corresponds to the phase of decreasing water cut and high oil production rate in the medium and high-permeability layers. When the cumulative injection volume reached 0.9 PV, the oil bank in the low-permeability layer also reached the core outlet, achieving a maximum length of 18.3 cm. Up to this point, the continuous migration, development, and production of the oil banks corresponded to the non-linear increase in the pressure curve.

After the injection volume exceeded 1.01 PV, the polymer front reached the outlet of the high-permeability core, corresponding to a sharp drop in the pressure curve. Similarly, the breakthrough of polymer in the medium-permeability layer also caused a significant pressure change.

Throughout the entire displacement process, the polymer front in the low-permeability layer never reached the core outlet, leaving a remaining oil bank length of 6 cm, accounting for one-fifth of the total core length. Due to the failure to control the fractional flow in the medium and high-permeability layers, the experiment had already entered a high water-cut state. Even continuing displacement with the water would make it challenging to fully produce the oil bank from the low-permeability layer.

5. Conclusion

(1) For heterogeneous reservoirs, based on the difference in water flooding development between layers, we analyzed and discussed 9 potential scenarios that may occur during chemical flooding, accounting for the changing positions of the water front, oil bank, and polymer front.

(2) Based on experiments and considering polymer properties like adsorption/retention and shear thinning, as well as the impact of the residual resistance of preceding slugs on subsequent slugs, this paper proposed a calculation method for flow resistance in multi-slug polymer flow. The characterization accuracy for fractional flow and pressure exceeded 96%.

(3) Based on oil–water relative permeability test experiments and polymer flow experiments, and considering the water flooding stage of multi-layer reservoirs and the oil bank migration and development patterns during polymer flooding, this paper proposed a flow resistance calculation method for multi-layer reservoirs under multi-slug polymer flooding using the superposition calculation of four parts: polymer, oil bank, oil–water two-phase, and pure oil phase. The characterization accuracy R^2 for fractional flow and pressure exceeded 97%.

(4) The results of the flow resistance algorithm indicate that in the multi-core parallel flow experiment, when the comprehensive water cut reached 98%, a 6 cm (one-fifth of the total core length) oil bank remained unproduced in the low-permeability layer. It is necessary to base slug selection and switching timing design on the internal fluid distribution within the cores and use interlayer flow resistance as guidance, thereby achieving effective mobilization of the low-permeability layer.

CRediT authorship contribution statement

Yaqian Zhang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Yiqiang Li:** Resources, Project administration, Funding acquisition. **Jian Zhang:** Resources, Formal analysis, Conceptualization. **Jinxin Cao:** Writing – review & editing, Validation, Formal analysis. **Xianjie Li:** Validation, Investigation, Formal analysis. **Cuo Guan:** Investigation, Visualization. **Chuanhui Miao:** Writing – review & editing, Validation. **Shilin Zeng:** Validation, Investigation. **Ke Hu:** Validation, Formal analysis. **Zheyu Liu:** Visualization, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Schroeter, S.A., Orme, A.M., Lehmann, K., et al., 2025. Hydroclimatic extremes threaten groundwater quality and stability. *Nat. Commun.* 16 (1), 9.
- Cui, B.H., Zhang, L.J., Wang, W.Q., et al., 2023. SPH approach for stability analysis of soil slope with variable permeabilities. *Soils Found.* 63 (4), 14.
- Rota, A., Possenti, L., Offeddu, G.S., et al., 2023. A three-dimensional method for morphological analysis and flow velocity estimation in microvasculature on-a-chip. *Bioeng. Transl. Med.* 8 (5), 12.
- Tusek, A.J., Salic, A., Valinger, D., et al., 2021. The power of microsystem technology in the food industry - going small makes it better. *Innov. Food Sci. Emerg. Technol.* 68, 18.
- Sachdev, S., Maugi, R., Davis, S., et al., 2020. Droplet factories: Synthesis and assembly of metal nanoparticles on magnetic supports. *J. Colloid Interface Sci.* 569, 204–210.
- Alfazazi, U., Thomas, N.C., Al-Shalabi, E.W., et al., 2024. Investigation of the effect of Residual Oil and Wettability on Sulfonated Polymer Retention in Carbonate under High-Salinity Conditions. *SPE J.* 29 (02), 1091–1109.
- Chen, C., Wang, S., Harwell, J.H., et al., 2021. Polymer-free viscoelastic fluid for improved oil recovery. *Fuel* 292, 120331.
- Bazarin, R.L.M., dos Santos, L.O.E., Siebert, D.N., 2025. Lattice Boltzmann method assessment for the Buckley–Leverett equation: Extension to dynamic capillary effect. *Phys. Fluids* 37 (5), 053328.
- Cao J, Li Y, Zhang Y, et al. Identification of Polymer Flooding Flow Channels and Characterization of Oil Recovery Factor Based On U-Net [Z]. *SPE Conference at Oman Petroleum & Energy Show*. 2024: D011S05R05.10.2118/218767-MS.
- Cao, J.X., Li, Y.Q., Zhang, Y.Q., et al., 2025. Viscous and capillary forces competition triggered flow instability in porous media. *Phys. Fluids* 37 (8), 14.
- Li, K., Du, X.L., Li, J., et al., 2025. A Novel Water-Flooding Characteristic Curve based on Fractal Theory and its Application. *Energies* 18 (6), 15.
- Wang, J.L., Song, H.Q., Wang, Y.H., 2020. Investigation on the micro-flow mechanism of enhanced oil recovery by low-salinity water flooding in carbonate reservoir. *Fuel* 266, 6.
- Xue, L., Zhang, J., Chen, G., et al., 2025. Enhance oil recovery by carbonized water flooding in tight oil reservoirs. *Geoenergy Sci. Eng.* 244, 12.
- Arab, D., Kantzas, A., Bryant, S.L., 2020. Water flooding of oil reservoirs: effect of oil viscosity and injection velocity on the interplay between capillary and viscous forces. *J. Petrol. Sci. Eng.* 186, 19.
- Wang, J., Qi, X.S., Hui, H.Q., et al., 2022. Mechanisms of remaining oil formation by water flooding and enhanced oil recovery by reversing water injection in fractured-vuggy reservoirs. *Petroleum Explor Dev* 49 (5), 1110–1125.
- Zhang, J., Li, Y., Li, X., et al., 2024. Mechanism of enhanced oil recovery by discontinuous chemical flooding in Bohai oilfield. *Acta Pet. Sin.* 45 (6), 988–998.
- Chen, Q., Ye, M., Wang, D., et al., 2024. Active polymer flooding for strongly heterogeneous reservoir associate with foam combination technology. *Geoenergy Sci. Eng.* 242, 213288.
- Li, X.J., Zhang, J., Zhang, Y.Q., et al., 2025. Experimental Study on the Application of Polymer Agents in Offshore Oil Fields: Optimization Design for Enhanced Oil Recovery. *Polymers* 17 (2), 20.
- Li, Y.Y., Liu, C.L., Zhang, Y.Q., et al., 2025. Experimental Investigation on the Application of Polymer Agents in Offshore Sandstone Reservoirs: Optimization Design for Enhanced Oil Recovery. *Polymers* 17 (5), 16.
- Beteta, A., Sorbie, K., Skauge, A., et al., 2025. Viscous crossflow as the mechanism for polymer enhanced oil recovery in viscous oils: evidence of crossflow between bypassed oil and viscous fingers. *Geoenergy Sci. Eng.* 253, 213968.
- Lu, X.G., Cao, B., Xie, K., et al., 2021. Enhanced oil recovery mechanisms of polymer flooding in a heterogeneous oil reservoir. *Petroleum Explor Dev* 48 (1), 169–178.
- Li, P., Shi, L., Wang, X., et al., 2025. Research on the “multi branch relay” chemical flooding mode for improving oil recovery in high water content and high heterogeneity reservoirs. *Geoenergy Sci. Eng.* 252, 213951.
- Chen, X., Trivedi, J., Liu, J.B., et al., 2025. Evolution law and control method of invasion front surface during the immiscible displacement in porous media: Expanding the swept volume. *Adv. Colloid Interface Sci.* 343, 34.
- Liu, Z., Cao, J., Wang, W., et al., 2025. Mobilization of Oil through In Situ Emulsification in a Porous Medium: a Comparative Analysis of Oil Solubilization and Interfacial Tension Reduction. *Langmuir* 41 (45), 30609–30620.
- Ahangar Darabi, K., Ahmadiouydarab, M., 2024. Injection of polyacrylamide and polyethylene glycol solutions to assess enhanced oil recovery and drawbacks of the polymer injection. *Geoenergy Sci. Eng.* 240, 213074.
- Dong, Z., Pan, X., Li, W., et al., 2024. Using polymer-alternating-water to maximize polymer flooding performance. *J. Pet. Explor. Prod. Technol.* 14 (6), 1589–1604.
- Hwang J, Sharma M, Chiotoroiu M-M, et al. Viscoelastic Polymer Injectivity: A Novel Semi-Analytical Simulation Approach and Impact of Induced Fractures and Horizontal Wells. *SPE Annual Technical Conference and Exhibition*. 2019: D021S3R06.10.2118/195978-MS.
- Mohammadnejad, H., Marion, B.A., Kmetz, A.A., et al., 2021. Development and experimental evaluation of a mathematical model to predict polymer-enhanced nanoparticle mobility in heterogeneous formations. *Environ. Sci. Nano* 8 (2), 470–484.
- Wang, Y., Yang, S.-L., Xie, H., et al., 2025. Dynamic characterization of viscoelasticity during polymer flooding: a two-phase numerical well test model and field study. *Pet. Sci.* 22 (6), 2493–2501.
- Chen, X., Patel, V., Liu, J.B., et al., 2025. Surfactant synergy on rheological properties, injectivity, and enhanced oil recovery of viscoelastic polymers. *Phys. Fluids* 37 (2), 13.
- Song, H., Ghosh, P., Bowers, A., et al., 2024. Low-Salinity Polymer Flood for Enhanced Oil Recovery in Low-Permeability Carbonates. *SPE J.* 29 (01), 443–454.
- Jin, F., Jiang, T., Wang, S., et al., 2024. Study on the channeling limit of polymer flooding in the conglomerate reservoir. *Geoenergy Sci. Eng.* 234, 212660.
- Lai, N., Wen, Y., Yang, Z., et al., 2020. Polymer flooding in high-temperature and high-salinity heterogeneous reservoir by using diutan gum [J]. *J. Petrol. Sci. Eng.* 188, 106902.
- Fakher, S., 2020. Development of novel mathematical models for laboratory studies of hydrolyzed polyacrylamide polymer injectivity in high-permeability conduits. *J. Pet. Explor. Prod. Technol.* 10 (5), 2035–2043.
- Xie, C., Xu, K., Qi, P., et al., 2022. A new mechanism of viscoelastic fluid for enhanced oil recovery: Viscoelastic oscillation [J]. *Advances in Geo-Energy Research* 6 (3), 267–268.
- Wu, D.J., Zhou, K., Zhao, F.J., et al., 2022. Determination of Permeability Contrast Limits for applying Polymer Solutions and Viscoelastic Particle Suspensions in Heterogeneous Reservoirs. *Energy Fuels* 36 (14), 7495–7506.
- Wu, D., Zhou, K., Hou, J., et al., 2020. Experimental study on combining heterogeneous phase composite flooding and streamline adjustment to improve oil recovery in heterogeneous reservoirs. *J. Petrol. Sci. Eng.* 194, 107478.
- Zhong, H., Shi, B., Bi, Y., et al., 2025. Interaction of elasticity and wettability on enhanced oil recovery in viscoelastic polymer flooding: a case study on oil droplet. *Geoenergy Sci. Eng.* 250, 213827.
- David, E., McDougall, S.R., Mackay, E.J., et al., 2025. Polymer Flooding Dynamics in Enhanced Oil Recovery: a Pore-Scale Study of the Influence of Shear-Thinning Rheology on Flow Dynamics and Recovery Efficiency. *J. Appl. Polym. Sci.* 142 (29), 18.
- Mansour, A., Al-Maamari, R.S., Souayah, M., et al., 2025. Experimental investigation and modeling of water alternating polymer flooding in homogeneous sandstone reservoirs. *Geoenergy Sci. Eng.* 247, 213725.
- W-LAKE L. Enhanced Oil Recovery [M]. 1989.
- Sun L, Li B, Jiang H, et al. An Injectivity Evaluation Model of Polymer Flooding in Offshore Multilayer Reservoir [J/OL]. 2019, 12(8):10.3390/en12081444.
- Cheng, J., Zhou, Q., Zhou, W., et al., 2020. Plugging method of polymer gel with controllable low initial viscosity for dominant flow paths in deep reservoirs and its application. *Acta Pet. Sin.* 41 (8), 969–978.
- Shi, L., Zhu, S., Cao, J., et al., 2019. Contribution degree of fluidity control for polymer flooding in heavy oil reservoir. *IOP Conf. Ser.: Earth Environ. Sci.* 237 (4), 042021.
- Cao, J., Zhu, S.J., Shu, Z., et al., 2022. Effects of residual resistance factor in the mobility control of the polymer flooding. *J. Appl. Polym. Sci.* 139 (48), 17.
- Zhong, H., Liu, H., Cui, T., et al., 2020. Numerical simulations of polymer flooding process in porous media on distributed-memory parallel computers. *J. Comput. Phys.* 400, 108995.